

Overview

This document summarizes the architecture, dataset, training strategy, and key findings of the micro-expression classification model built using the CASME II dataset as part of the dissertation project on liveness verification in financial KYC.

Emotion Categories Captured

- 😊 **Positive** (e.g., happiness)
 - 😞 **Negative** (e.g., disgust, repression)
 - 😲 **Surprise**
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Model Architecture

- **Model Type:** CNN + LSTM (Temporal)
 - **Base Feature Extractor:** TimeDistributed `MobileNetV2` (frozen, pretrained on ImageNet)
 - **Temporal Component:** `LSTM(128)`
 - **Classifier Layers:**
 - `Dense(64, activation='relu')`
 - `Dense(3, activation='softmax')`
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Training Setup

- **Input Shape:** (16, 224, 224, 3)
 - **Training Epochs:** 30 (Early stopping triggered at epoch 23)
 - **Loss:** Categorical Crossentropy
 - **Optimizer:** Adam (learning rate = 1e-4)
 - **Enhancements:**
 - Class weight balancing 🙌
 - Early stopping with best weight restore 🙌
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Final Results

Model trained on 132 video sequences (16 frames each) and validated on 27 sequences

Emotion	Precision	Recall	F1-score	Support
Positive	0.50	0.50	0.50	6
Negative	0.87	0.81	0.84	16
Surprise	0.67	0.80	0.73	5

Emotion	Precision	Recall	F1-score	Support
Accuracy			74%	27

👉 Model now predicts all classes including `positive` (which was previously missed).

Key Insights

- Class weights significantly improved performance on the `positive` class.
- Model is now balanced across all three emotion categories.
- Best validation achieved at epoch 18.
- Suitable for integration with deepfake detection model in a late-fusion liveness verification setup.

📁 Output Assets

- Trained model (Keras): pending save
- Evaluation script: `train_model.py`
- Preprocessed data source: `microexpression_processed/`
- Sequence files: `X_sequences.npy`, `y_labels.npy`

📅 Date Logged

June 22, 2025